



Semiconductor Industry in Brazil

*Changing Semi Conductor Dynamics
The Brazilian Landscape*

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EXECUTIVE SUMMARY

A 7-fold strategy to revive Semiconductor Co. and reform industry, changing the chip landscape of Brazil

Company-Specific solution

	Complication	Proposed Solution	Parameters of Assessment				Overall Attractiveness
			Investment Required (CapEx: Small/Medium/Large)	Degree of Impact	Complexity of Implementation	Duration of Implementation (Rankings 1-7: Longest - Fastest)	
1	Reversing Liquidation: Business Strategy	Becoming Fabless	Small	Utilisation of DH & Job Creation	Medium	7 th (Least Time)	
2	Reversing Liquidation: Business Strategy	Outsourcing to Foundries	Medium	26.95% Cost Saving	High	5 th	
3	Industry Positioning to Gain Competitiveness	Sectoral Diversification	Medium	2 New Industries Catered	Medium	2 nd	
4	Industry Positioning for Competitiveness	Pricing	Small	3.2% → 5% Market share	Low	6 th	
5	Building Cross Collaboration	Employee Retention	Small	74% Increase in Job Satisfaction	Low	1 st (Most Time)	
6	Semiconductor Industrial Reform	Policy Reforms	Large	5.8 Billion USD Investment Outlay	Medium	4 th	
7	Semiconductor Industrial Reform	Transformation into OSAT hub	Large	4+ Trade Agreements	High	3 rd	



Sources: Team Analysis

1) THE NEED FOR TRANSFORMING FROM FAB TO FABLESS

Changing industry dynamics require expertise in R&D with open innovation and joint development projects

Economic Profit Value Creation(2015-19, Billion USD)

Cumulative profit of \$335 billion with top companies that drive the value

Memory	> Micron, Samsung, Western Digital	94.9
Fabless Semiconductor	> Apple, Broadcom, Nvidia, Qualcomm	81.6
Microprocessing unit (MPU)	> Intel	44.4
Capital equipment	> AMAT, ASML, Tokyo Electron	41.2
Foundry	> TSMC, DB HiTek, Tower Semiconductor	35.8
IDM	> Infineon, NXP, Renesas	27.1
Others		15.9



Huge Capacity in R&D with Expertise in IC design and Memory- Related Products, such as eMMCs, eMCPs, SSDs

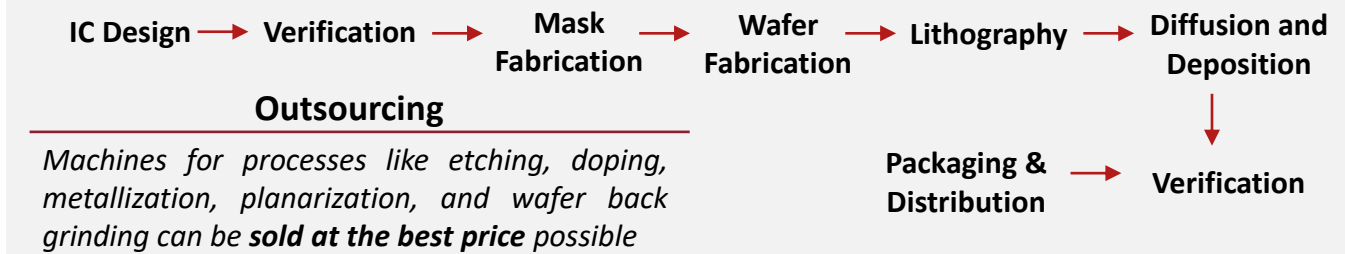


22
Design Houses across Brazil
134,000
Potential Electrical And
Computer Science Engineers

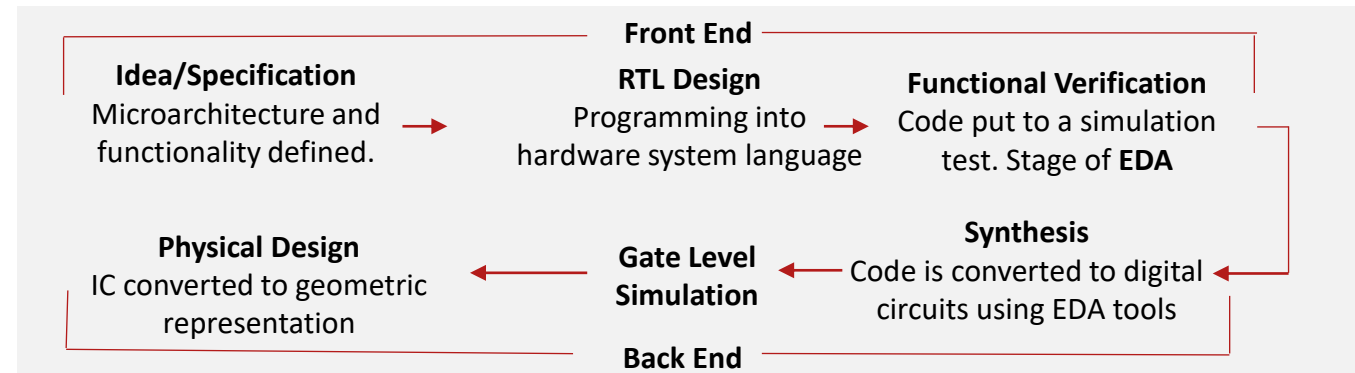
Sources: McKinsey, PWC, KPMG

Transforming from IDM to Fabless (Moving Away from Manufacturing)

Strategic Planning



Technical Expertise



Intellectual Property Planning

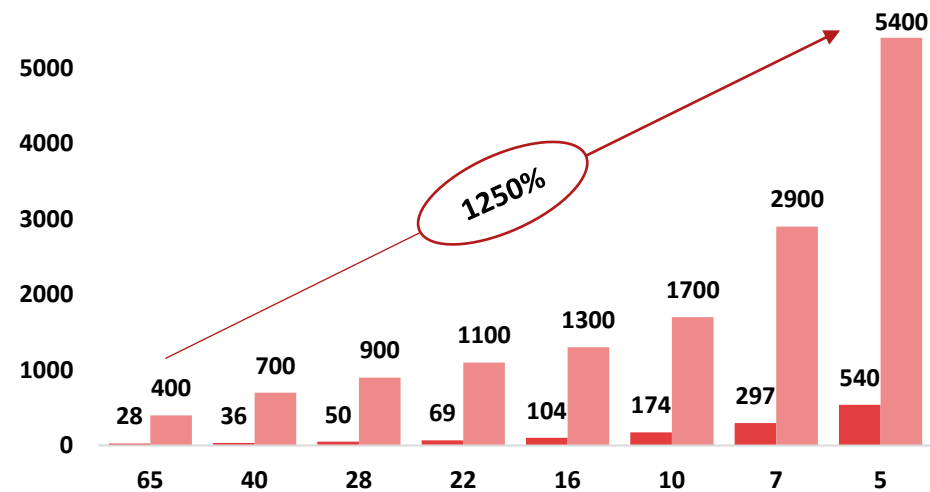
- ✓ Estimated worth of Semiconductor Intellectual Property Market - **\$6 billion (2022)**
- ✓ Need to adopt **patenting process, NDA agreements and copyrights** due to increased competition

2) OUTSOURCING TO FOUNDRIES

Data-driven Decisions, Analytics and AI can transform Indigenous Fabs but at present, handing out is the key

The Relationship between Chip Sizes and Concerned Costs

As chip sizes get smaller from 65nm to 5nm for greater precision, the manufacturing costs increase many fold compared to R&D cost



The Capital Intensive Foundry Business

- ✓ Soaring Labour Costs
- ✓ Stiff Competition



3 Billion USD
Cost of Setting up a Single
Fabrication Plant

26.95%
Cost that can be Saved By
Outsourcing the Process

Sources: Accenture, Semi-engineering, Smart Modular

Increased ROI by Dropping Core Competencies in Microfabrication and Wafer Testing

Establishing Tie-ups with Brazilian Foundries and MNC

Tier 1 Suppliers:
Brazil Originated



Tier 2 Suppliers:
MNCs having fabs in Brazil



Tier 3 Suppliers:
Importing from Taiwan



Building a Resilient Supply Chain

Problems and Identified Choke Points

- ✓ **Diverse Demand Patterns** - Shift from build to stock supply system because of innovation
- ✓ **Geopolitical Instability**- Semiconductors cross border 70 or more times)
- ✓ **Fragmented Product Supply** - Different wafer sizes and 100 days of development cycle)

Solution: Achieving **Automation** and **Modernization** over **Cycle Time** (Industry Driver)

Automation
Setup



Equipment
interface

Manufacturing
Execution System

Real-Time
Dispatch for WIP

Recipe
Server

Equipment
Interface



Manufacturing
Request (FSC)

Requirement
Analysis (SECS/GEM)

Design, Development,
Testing & Production

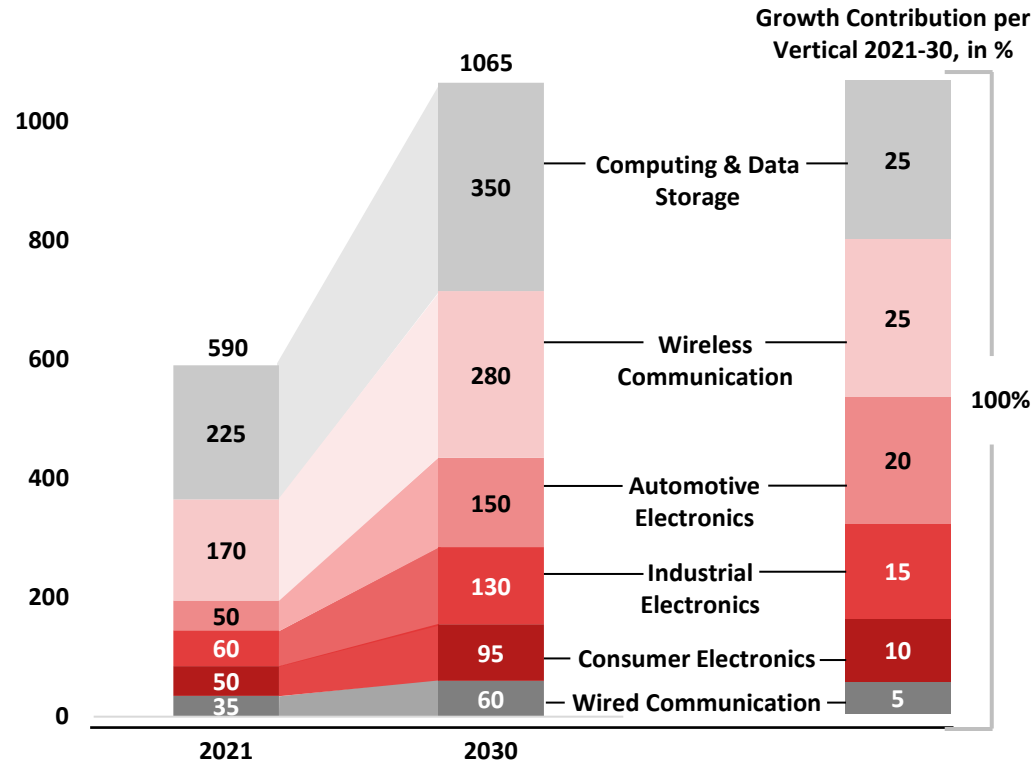
Linking Platform for Fab Systems: **MES, System setup, FDC, SPC, and APC**

3) REVIVAL THROUGH SECTORAL DIVERSIFICATION

Entering high yielding fields of Computing, Data Storage and Wireless Communication

Global Semiconductor Market Value by Vertical (Billion USD)

Driven by Automotive, Data Storage and Wireless Industries



“ Semiconductor Co. needs to shift focus from agriculture and industrial electronics towards **Research and Development of Computing, Data Storage And Wireless Communication Chips** ”

Sources: Mckinsey, Fortune Business Inside

Factors

Computing and Data Storage

Wireless Communication

Sector Overview

Growing at a **CAGR of 7.1%**, the industry focuses on management of digital files and algorithmic processes

Growing at a **CAGR of 7.35%** the telecom, Broadcom and fixed wireless network is expected to boom

Demand Vs Supply

Incoming of **AI, ML, and IoT** the demand will skyrocket but the supply is growing at a **lower rate of 5.46%**

There would be an **increase of 1.42 billion devices** by 2025 and the supply for chips will go up

Technical Requirements

• GPU • ASICs } Used via **SoC**
• FPGA • CPU } (System on a Chip)

• NFC • UMTS
• SPI • CDMA

Profitability

CP: 400 USD/Chip on average; SP:500
Exceptional Creation: Nvidia's A100 GPU is priced at 10,000 USD

Relatively Low Margin as compared to computing but **better returns** than agriculture

Other Requirements

Optimal Processors, Memory Arrays, Security, and Real-time Data Connectivity

Performance Specifications include Supply Voltage, Data Rate, and Temperature Junction

Competition



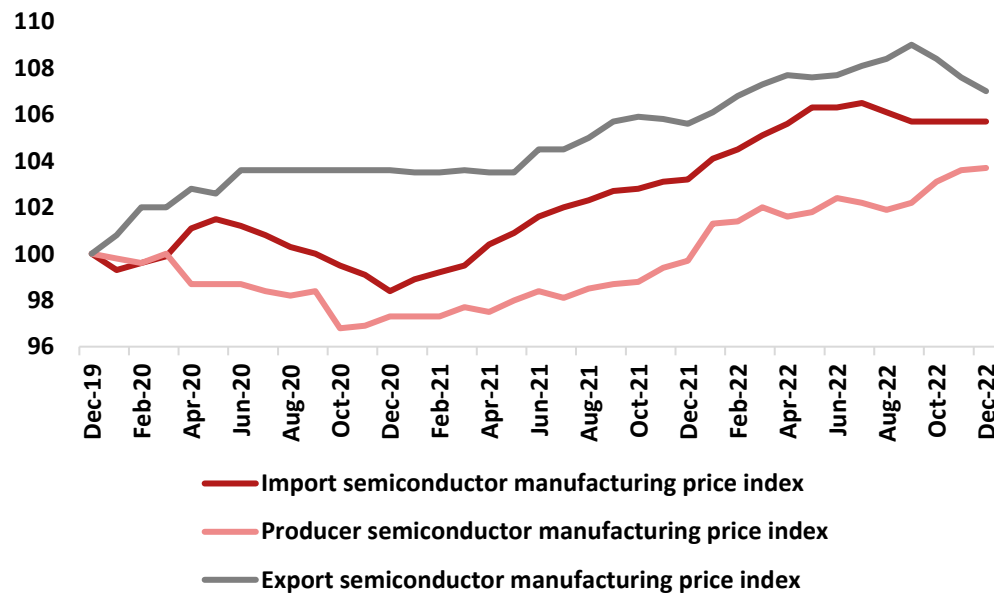
Present Synergies with Company



4) DEVISING THE OPTIMAL PRICING

Comparing industry – wide competitors pricing and devising optimal cost structure for the company

Import, Export, And Producer Semiconductor Manufacturing Price Indices by US Labor Statistics (Price Of Dec 2019)



Competitor Investments in the Industry



- ✓ **25% price hike** over their current 5nm wafer; the new wafers with 3nm node technology will be priced at **\$20,000 a piece**



- ✓ It costs Nvidia around **\$3,320** to manufacture an H100 GPU. With retail prices averaging between **\$25,000** and **\$40,000**, this represents a **profit of up to 1000%**

Sources: Computer world. Mint, The Economist

Key Factors and Proposed Price Strategy for Semiconductor Co.



Research and Development Cost

The jump in R&D costs from improving a 10nm minimum feature size on chips to a 7nm minimum feature size is **\$100 million**

Economic Factors

The cost of making semiconductor chips can be influenced by various economic factors, such as **inflation** and **currency exchange rates**



Labor Cost

Although it varies by country; in the US, the average **annual salary** for a semiconductor engineer is **\$118,300**, while in Taiwan, it is **NT \$980,000**

Final Pricing

70 USD for basic chips, **4000 USD** for advanced chips, and **10000 USD** for specialized chips



5) EMPLOYEE ACQUISITION AND RETENTION STRATEGY

Targeting Industry Pain Points through a 3 fold HireHarbor Strategy

Key Issues



Talent Shortage

1 Problem of the Semiconductor Industry

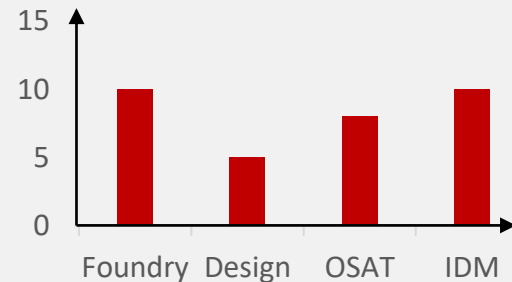
Reasons Identified

- Limited Global Visibility** compared to international counterparts like Intel, TSMC, Qualcomm
- Resource Constraints** in terms of cutting-edge technology and budget
- Political Instability** and Market Volatility of the Brazilian Economy
- Low Average Employee Satisfaction Score** compared to the industry average

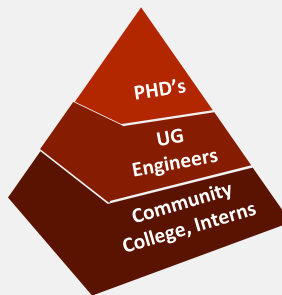


Project HireHarbor

HireHarbor: A Retention Revolution



Processes that can be Automated per Segment



The Supply – Demand Gap for Talent

Boost competitiveness by training new graduates for technical and process engineering roles



Talent Win Room Strategy

Talent Acquisition, Attraction, and Selection

- ✓ Brand Building & Awareness
- ✓ External Full-Time Recruiting
- ✓ Internal-Transfer Recruiting
- ✓ Contracting Agencies

New Hire Experience & Progression

- ✓ Onboarding and Dynamic Deployment
- ✓ Compensation, Career Paths & Job Structure

Why Automation?

- ➔ Optimises Employee Time that can be devoted for **Personal Growth**
- ➔ Tackles talent acquisition, training and retention costs amid a **Cross-sector Talent War**

Implementation Ideas...

1. Accelerating Career Growth

- ✓ Early Career Pivotal Learning : 10-month programs for employees to learn new technical skills

2. Nurturing Talent from the start

- ✓ Collaborating with educational institutes for semiconductor trainings

3. Immersive Experiences Programs

- ✓ Building relationships with fellow STEM enthusiasts and exposing them to the Semiconductor Industry on par with Big 4 tech and automotive companies

4. Greater Growth Opportunities

- ✓ Empowering new hires to rotate across functions to build diverse skills including core technical and business skills

Sources: Mckinsey, Accenture, KPMG, Handshake Blog

6) GOVERNMENTAL INITIATIVES TO THE RESCUE

2 – pronged approach of incentivizing domestic production with international collaboration to optimize 5.8 Billion outlay

Incentivising Domestic Production of Fabless and OSAT Industry

Pain Points of the Brazilian Semiconductor Industry...



Placing Brazil on the Global Forefront via Project RioTech Genesis...



- Production Linked Incentives:** Fiscal incentives amounting to **50% of project cost** can be made available to companies for setting up Fabless or OSAT units with a **tax break at 25%**
- Design Linked Incentives:** Financial Incentives and Design Infrastructure support Integrated Circuits (ICs), Chipsets, System on Chips (SoCs), Systems and IP Cores and semiconductor linked design over a period of **5 years**
- Upskilling:** Setting up courses in public universities to specifically train workers

Sources: ASML, European Commission

Policy Plans for International Collaborations and the Shift towards Industry Growth on the Global Stage

Leveraging International Partnerships...

Free Trade Agreement with European Union

Importing EUV machines from ASML for manufacturing the most advanced chips

Collaborating with Taiwan

Learning technical know-how and building preferential FDI tariffs for OSAT setup with Foxconn

China as the Raw Material Supplier

Import of critical minerals such as gallium, germanium and raw silicon along with R&D for improved technology

Fabless Manufacturing with USA

USA can outsource Chips production to Brazil at lower costs and supply advanced expertise for IC Design

Investment Outlay

5.8 Billion USD

Annual Investment under Project RioTech Genesis

Investment Breakdown

Tax Benefits

Workforce Development

Design Linked Incentives

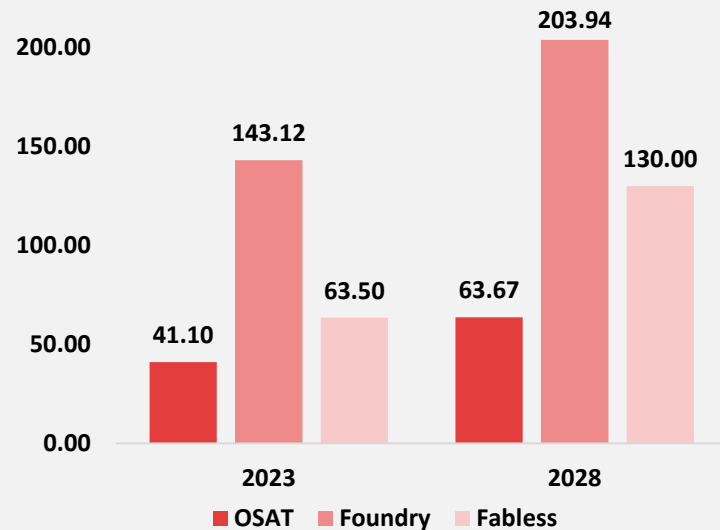
Production Linked Incentives

7) TRANSFORMATION INTO OSAT HUB

Overcoming Low Profit Margin through Mass Scaling of OSAT industry in Brazil Via Public-private Partnership

Understanding Outstation Semiconductor Assembly and Testing

Expected Growth in the Market in Billion USD



Key Competitors having both Foundry and Packaging Business

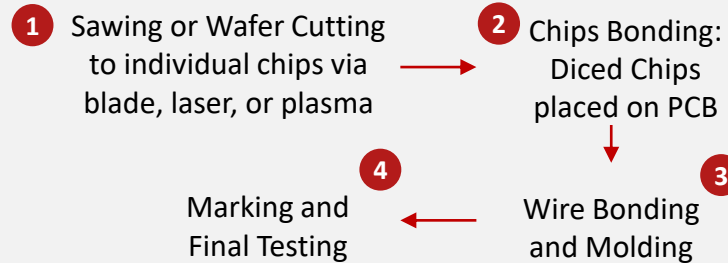
CAGR till 2028
 OSAT- 9.15%
 Foundry- 7.34%
 Fables- 6.25%

Profit Margins
 OSAT- 25%
 Foundry- 45%
 Fables- 45%

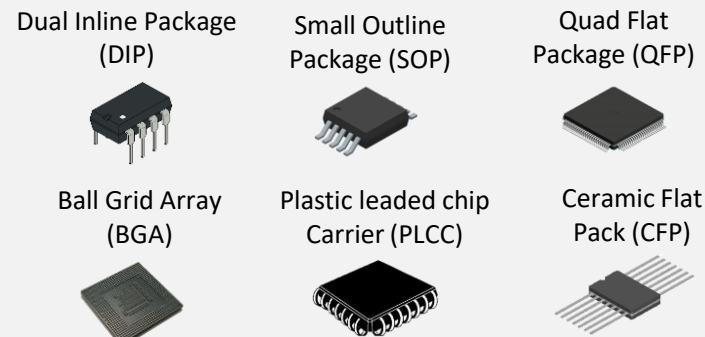


Technical Requirements

The Process...

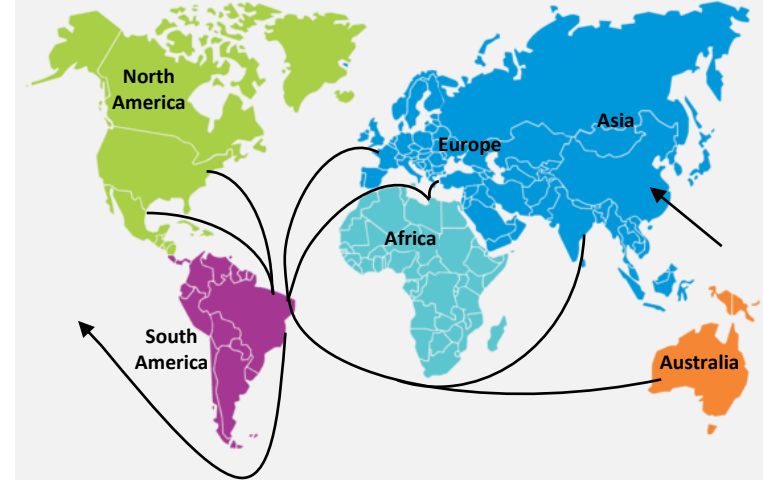


Types of Packaging



- ✓ Valued at **USD 37.22 billion USD** in 2022
- ✓ **Cheaper** than Foundry Business
- ✓ **Asian Market Domination** (Taiwan, China, Korea)

How can Brazil become an OSAT Hub?



Growth Triggers

- ✓ **Access to Major Economies** - Easy trade routes and developed air transportation system
- ✓ **Favourable Economy and Demographics** - 10th largest GDP and ranks 7th as per population
- ✓ **Cheap Labour** - Ranked 79 in per capita income

Government Initiatives

- ✓ Production-linked Incentive Schemes
- ✓ Tax Benefits
- ✓ Low Import Duties On Raw Material

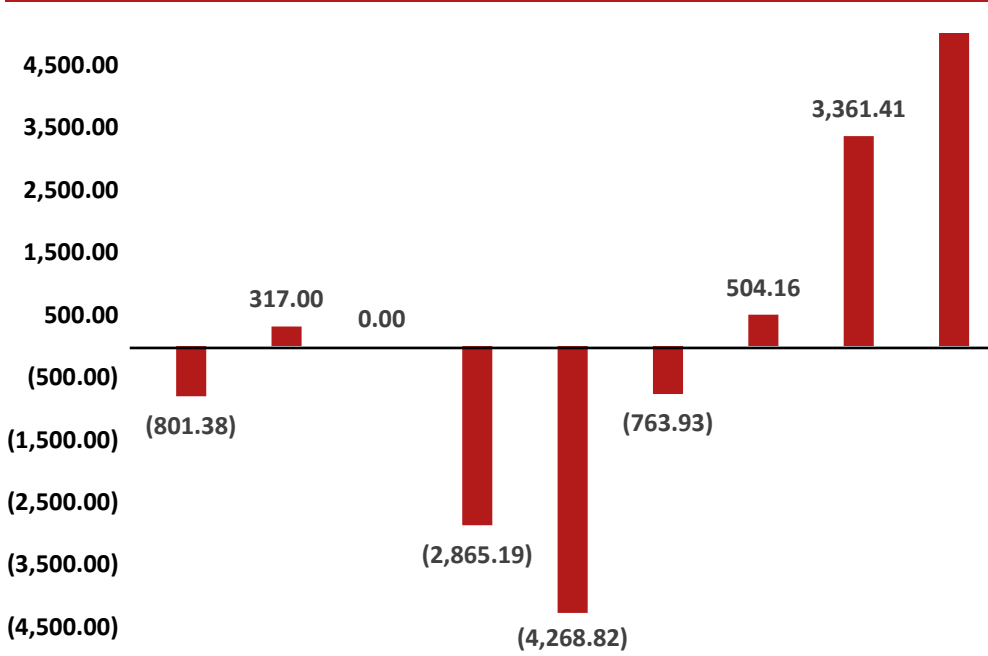


Sources: KBV Research, straits research, grand view research

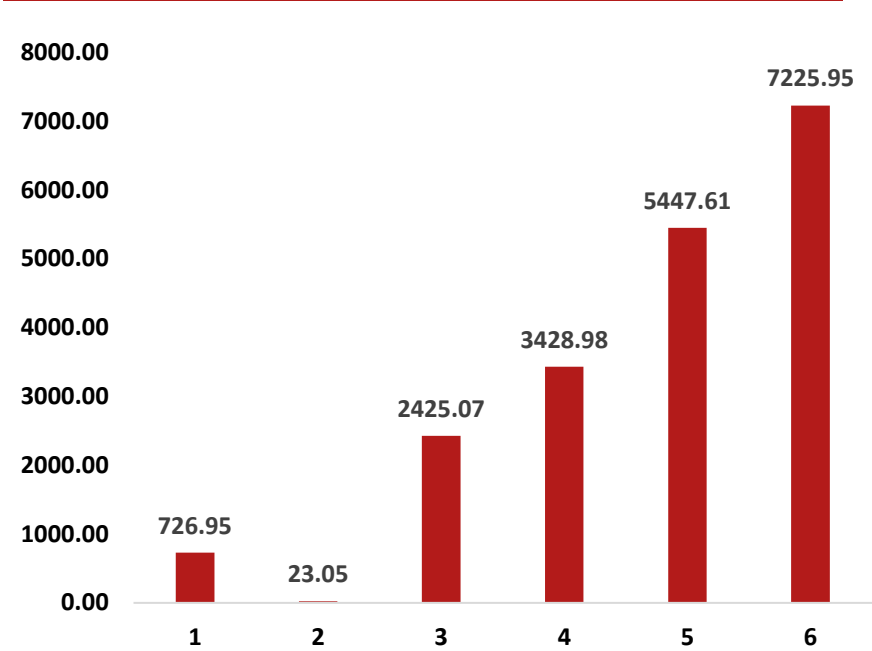
FINANCIAL ANALYSIS

A positive NPV and IRR of 16% makes the entire sequence of the project highly viable and implementable

Net loss to net gains from 2020 to 2028. Semiconductor Co. business model revamp



The growth rate in cash flows from 2022 to 2028



IRR of 16%
Making the Project
Highly Feasible

**\$13,800
Thousand**
Net Present Value for
project spanning 6 years

44%
Estimated increase in
Revenue in the next 6 years

The **viability** and **impact** of Semiconductor.Co's operations after adopting the suggested strategies can be measured and monitored using **key performance indicators (KPI)**

Measuring and Monitoring the Impacts of Project KPIs

1
R&D Spending as a percentage of Revenue
If increase research investment is increasing revenue should also increase making this ratio constant

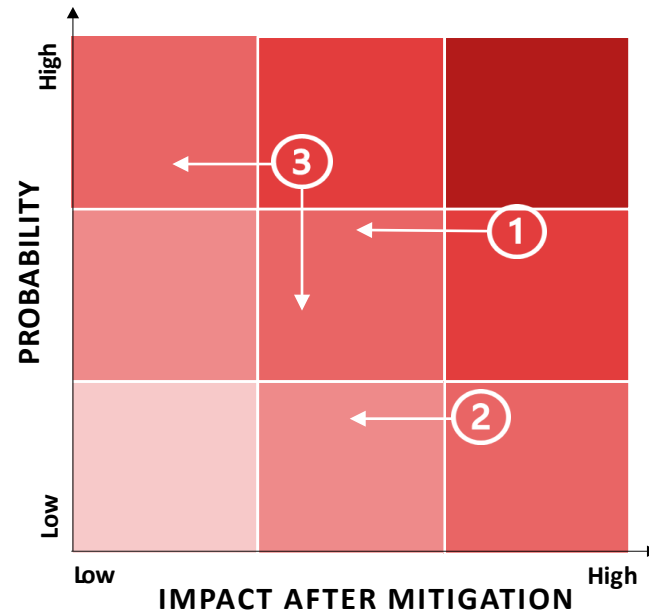
2
Cycle Time
This is the total time from the beginning to the end of a process, as defined by the company and its customers. It measures the efficiency of their production processes

3
Foreign Direct Investment
With increasing efforts towards trade policies and capital investment potentials collabs should increase, raising this metric

Sources: Team Analysis

RISK MITIGATION & TIMELINE

Analyzing the potential roadblocks and strategies that can develop the Semiconductor industry by 2030



①

High Initial Investment

Risks

Shifting to Fabless segment will require heavy investment in R&D and setting up of OSAT's

Mitigation

Disinvestment in Foundries segment will release investment for in Fabless and OSAT segment

②

Uncertainty around Success of Partnerships

Differing goals and coordination of partners may hamper effectiveness of partnerships

Thorough due diligence, clear communication, adaptability, and regular performance evaluation partnerships will be a success

③

Demand Supply Imbalance

Shortages of Raw Materials, supply chain inefficiencies, long lead times may make it difficult to respond to sudden shifts in demand

Diversified Supplier Partnerships and Strategic Alliances will provide a safety net for quick action in case of any sudden shift in demand

PROJECTED TIMELINE

2023

2024

2025

2026

2027

2028

2029

Transformation to a Fabless Company

Outsourcing to Foundries

Sectoral Diversification

Revaluating Pricing

Building Cross Collaborations

Policy Reforms and making Brazil OSAT hub

Sources: Team Analysis

Executive Summary

Business Strategy

Positioning

Cross Collaboration

Industrial Reform

Financials

Risk & Time Analysis

The background of the slide is a detailed, artistic rendering of a microchip and its surrounding circuitry. The microchip is a dark, square component with a grid of pins on its sides, positioned centrally. It is surrounded by a complex network of glowing blue and orange circuit lines that spread out across the frame. Numerous small, glowing orange and yellow dots are scattered throughout the circuitry, creating a sense of digital activity and energy. The overall color palette is dominated by cool blues and teals, contrasted with warm oranges and yellows from the glowing elements.

APPENDIX

Financials (1/5)

INCOME STATEMENT : SEMICONDUCTORCO									
Years	2020A	2021A	2022A	2023E	2024E	2025E	2026E	2027E	2028E
Gross Revenue									
Sale of chips	2,212.02	4,340.31	NA	4,611.58	4,899.80	5,206.04	5,531.42	5,877.13	6,244.45
Deductions									
Taxes	(448.72)	(906.48)	NA	(691.74)	(734.97)	(780.91)	(829.71)	(881.57)	(936.67)
Returns	0.00	(36.44)	NA	(36.44)	(36.44)	(36.44)	(36.44)	(36.44)	(36.44)
Net Sales	1,863.79	3,299.03	NA						
Cost of Goods Sold	(315.70)	(592.17)	NA	(593.31)	(594.43)	(595.52)	(596.59)	(597.64)	(598.69)
Gross Profit	1,444.14	2,703.99	NA	3,290.09	3,533.97	3,793.18	4,068.68	4,361.48	4,672.66
Operational Expenses									
Employees	(7,031.45)	(5,670.41)	(3,792.51)	(10,341.93)	(10,498.32)	(10,633.86)	(10,745.33)	(10,829.20)	(10,881.58)
General and Administrative	(5,663.54)	(5,582.27)	(5,718.38)	(8,869.20)	(7,953.59)	(6,888.88)	(5,660.00)	(4,250.62)	(2,642.94)
Directors	(416.38)	(182.01)	(148.52)	(530.72)	(384.68)	(503.93)	(484.85)	(542.02)	(561.62)
Other Operating Income/Expenses	11,120.34	9,922.76	7,583.51	16,863.21	14,559.51	17,253.23	17,383.94	18,973.86	19,892.28
Loss From Operations	(830.45)	(1,605.99)	0.00	(2,878.64)	(4,277.07)	(773.43)	493.76	3,352.02	5,806.14
Financial Result									
Financial Expenses	(52.75)	(24.05)	(9.46)	(28.75)	(20.75)	(19.66)	(23.05)	(21.15)	(21.29)
Financial Income	81.81	28.54	16.27	42.21	29.01	29.16	33.46	30.54	31.05
Net Loss	(801.38)	317.00	0.00	(2,865.19)	(4,268.82)	(763.93)	504.16	3,361.41	5,815.90
Average Number of Shares (in thousands)	8,085.00	7,539.00	7,946.00	7,946.00	7,946.00	7,946.00	7,946.00	7,946.00	7,946.00
Net Loss per Share	8.09	17.79	0.00	(0.36)	(0.54)	(0.10)	0.06	0.42	0.73

Financials (2/5)

CASHFLOW STATEMENT : SEMICONDUCTORCO									
Years	2020A	2021A	2022A	2023E	2024E	2025E	2026E	2027E	2028E
Cash flows from Operating Activities									
Net Income(loss)	(801.38)	317.00	(2,980.28)	(2,865.19)	(4,268.82)	(763.93)	504.16	3,361.41	5,815.90
Adjustment	308.00	3.59	145.31	152.30	100.40	132.67	128.46	120.51	127.21
Provision for Labour Contingencies	(24.45)	(730.03)	(199.42)	(317.97)	(415.81)	(311.06)	(348.28)	(358.38)	(339.24)
Labour Contingency Reversal	2,448.60	2,318.42	2,378.24	2,381.75	2,359.47	2,373.15	2,371.46	2,368.03	2,370.88
Depreciation	179.22	144.32	123.36	123.36	123.36	123.36	123.36	123.36	123.36
Amortization			4.35	4.35	4.35	4.35	4.35	4.35	4.35
Operating Assets and Liabilities									
Increase/Decrease in Customers	215.41	(826.06)	1,249.67	1,312.15	1,377.76	1,446.65	1,518.98	1,518.98	1,518.98
Increase/Decrease in Inventories	(108.19)	367.62	(10.41)	83.01	87.16	91.51	96.09	100.90	105.94
Increase/Decrease in Recoverable Taxes	(91.63)	253.45	(514.25)	(117.48)	(126.09)	(252.61)	(165.39)	(181.36)	(199.79)
Increase/Decrease in Prepaid Expenses	137.45	97.83	(68.30)	55.66	28.40	5.25	29.77	21.14	18.72
Increase/Decrease in Judicial Deposits	(177.49)	(354.69)	(30.46)	(187.55)	(190.90)	(136.30)	(171.58)	(166.26)	(158.05)
Increase/Decrease in Advances to Employees and Suppliers	(97.41)	145.75	(30.46)	5.96	40.42	5.31	17.23	20.98	14.51
Increase/Decrease in Non-current Assets Recoverable Taxes	179.41	54.39	1,031.14	421.65	502.39	651.73	525.26	559.79	578.92
Increase/Decrease in Other Credits			(435.92)	NA	NA	NA	NA	NA	NA
Increase/Decrease in Suppliers	42.16	(100.52)	28.57	31.43	34.57	38.03	41.83	46.01	50.61
Increase/Decrease in Tax Obligations and Provisions	(3.08)	139.29	(144.74)	(2.84)	(2.76)	(50.12)	(18.57)	(23.82)	(30.84)
Increase/Decrease in National Treasury	196.93	(245.74)	452.19	134.46	141.18	148.24	155.65	163.44	171.61
Increase/Decrease in Labor Obligations and Provisions	199.62	(419.13)	55.25	(54.75)	(139.54)	(46.35)	(80.22)	(88.70)	(71.76)
Net Cash used in/provided by Operating Activities	2,603.17	1,165.49 	1,049.49	1,681.69	1,752.58	1,901.34	1,949.04	1,971.09	1,998.87

Financials (3/5)

BALANCE SHEET : SEMICONDUCTORCO								
Years	2020A	2021A	2022A	2023E	2024E	2025E	2026E	2027E
Assets								
Current Assets								
Cash and cash equivalents	564.22	3584.62	4829.33	2992.72	3802.22	3874.76	3556.57	3744.52
Short-term marketable securities	888.58	135.16	145.31	389.68	223.38	252.79	288.62	254.93
Accounts Receivable	385.58	1185.6	0	523.73	569.78	364.50	486.00	473.43
Inventory	3213.98	2629.32	2781.81	2875.04	2762.06	2806.30	2814.46	2794.27
Tax Credits	439.29	156.17	678.85	424.77	419.93	507.85	450.85	459.54
Employee and Supplier Advances	221.76	61.03	94.79	125.86	93.89	104.85	108.20	102.31
Financial Credits - PADIS	0	0	435.92	145.31	193.74	258.32	199.12	217.06
Prepaid Expenses	107.8	2.69	71.14	60.54	44.79	58.82	54.72	52.78
Total current assets	5801.95	7.75	9037.14	4948.95	4664.61	6216.90	5276.82	5386.11
Non-Current Assets								
Judicial Deposits	2455.34	2589.83	1729.1	2258.09	2192.34	2059.84	2170.09	2140.76
Tax Credits	446.99	771.49	843.64	687.37	767.50	766.17	740.35	758.01
Property, Plant and Equipment	2008.35	1818.34	885.46	1570.72	1424.84	1293.67	1429.74	1382.75
Intangible Assets	15635.43	12354.81	10655.37	12881.87	11964.02	11833.75	12226.55	12008.11
Total non-current Assets	2759.87	1209.83	1151.85	1707.18	1356.29	1405.11	1489.53	1416.97
Total Assets	20850.64	16154.46	13536.31	16847.14	15512.64	15298.69	15886.16	15565.83
	26652.59	23909.04	22573.45	48438.97	31640.48556	34217.63407	38099.02877	34652.3828

Financials (4/5)

Liabilities

Current Liabilities								
National Treasury Advances	747.09	450.9	927.46	708.48	695.61	777.19	727.09	733.30
Suppliers	107.8	0	28.57	45.46	24.68	32.90	34.34	30.64
Tax Obligations and provisions	9.05	147.73	10.97	55.92	71.54	46.14	57.87	58.52
Labour Obligations and provisions	1065.87	574.76	661.06	767.23	667.68	698.66	711.19	692.51
Total Current Liabilities	1929.81	1173.39	1628.07	1577.09	1459.52	1554.89	1530.50	1514.97
Non-current Liabilities								
Contingencies	0	0	0	0.00	0.00	0.00	0.00	0.00
Contingencies	1487.83	660.92	642.52	930.42	744.62	772.52	815.86	777.67
Total Liabilities	3417.65	1834.31	2270.59	4084.60	2729.83	3028.34	3280.92	3013.03

Equity

Stockholder's Equity	24,312.75	22,670.85	23.90	23.90	23.90	23.90	23.90	23.90
Equity Valuation Adjustment	2,824.17	2,258.47	1,985.28	2,355.97	2,199.91	2,180.39	2,245.42	2,208.57
Advances for Future Capital Increase	1,405.83	1,402.79	1,489.76	1,432.79	1,441.78	1,454.78	1,443.12	1,446.56
Accumulated Deficit	(5,307.80)	(4,257.38)	(7,068.13)	(5,544.44)	(5,623.32)	(6,078.63)	(5,748.79)	(5,816.91)
Total Equity	23,234.94	22,074.73	20,302.86	21,870.84	21,416.14	21,196.62	21,494.53	21,369.10
Total Liabilities and equity	26652.59	23909.04	22573.45	24378.36	23620.28	23524.03	23840.89	23661.74

Financials (5/5)

Calculation of PV to FCFF	2023E	2024E	2025E	2026E	2027E	2028E
EBIT	1153.08	36.55	3846.59	5438.99	8640.90	11461.67
Tax Rate	15.00%	15.00%	15.00%	15.00%	15.00%	15.00%
EBIT(1-T)	980.12	31.07	3269.61	4623.14	7344.77	9742.42
Less: reinvestment rate	25.83%	25.83%	25.83%	25.83%	25.83%	25.83%
free cash flow to firm (FCFF)	726.95	23.05	2425.07	3428.98	5447.61	7225.95
mid year convention		0.5	1.5	2.5	3.5	4.5
discounting factor		0.9573169	0.8773384	0.8040417	0.7368685	0.6753073
present value of FCFF		22.06	2127.60	2757.05	4014.18	4879.74

expected growth	6.71%
terminal growth	2.38%
Cost of Equity	9.12%

Calculation of Terminal Value

FCFF (n+1)	7,710.81
WACC	9.12%
terminal growth rate	2.38%
terminal value	1,14,471.13

Calculation of equity value per share

PV of FCFF	13,800.62
terminal value	1,14,471.13

Cost of Equity

Risk-free Rate	4.18%
Implied ERP	3.29%
Country Risk Premium	1.36%
Beta	1.06
Cost of Equity	9.12%

Particulars	Investment	2023E	2024E	2025E	2026E	2027E	2028E
FCFF	(58,000.00)	726.95	23.05	2,425.07	3,428.98	5,447.61	7,225.95
Terminal Value							1,14,471.13
Total	(58,000.00)	726.95	23.05	2,425.07	3,428.98	5,447.61	1,21,697.08
IRR							16%